Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

683
-24 -3.39%
Sep 23rd, 2024

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

794
-25.00 -3.05%
Sep 23rd, 2024

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

610
-7 -1.13%
Sep 23rd, 2024

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

89.79
-2.06 -2.24%
Sep 23rd, 2024

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

100.04
-0.22 -0.22%
Sep 23rd, 2024

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

820
-20 -2.38%
Week Ending Sep 20th, 2024

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2501 (sep) RMB/t (3pm close)

658.50
-21.50 -3.16%
Sep 23rd, 2024 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
October 24 USD/dmt

89.45
-2.70 -2.93%
Sep 23rd, 2024 (5.30 pm Print)

SHFE Rebar
RB2501 (oct) RMB/t

3087
-78 -2.46%
Sep 23rd, 2024 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

28.04
-0.19 -0.67%
Sep 20th, 2024

C5, W. Australia - Qingdao USD/t

11.69
-0.20 -1.68%
Sep 20th, 2024

Steel Price

Steel Rebar (China Domestic) RMB/t

3310
0 0.00%
Week Ending Sep 20th, 2024

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

150.00
-0.72 -0.48%
Week Ending Sep 20th, 2024

Steel Inventory in China
million tonnes

9.58
-0.33 -3.30%
Week Ending Sep 20th, 2024

Steel Price

Steel HRC (China Domestic) RMB/t

3140
10 0.32%
Week Ending Sep 20th, 2024

IRON ORE PORT STOCK INDEX (IOPI)

Sep 23rd, 2024		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	683	-24.0	-3.39%	880	895	770	1063	89.33	-3.13	-3.4%	116.38	121.03	101.00	140.00
IOPI58	58% Fe Fines	610	-7.0	-1.13%	785	781	665	963	80.25	-0.84	-1.0%	104.50	106.10	86.00	128.00
IOPI65	65% Fe Fines	794	-25.0	-3.05%	996	1026	881	1175	104.47	-3.24	-3.0%	132.21	139.47	116.00	155.00

IRON ORE SEABORNE INDEX (IOSI)

Sep 23rd, 2024		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	89.79	-2.06	-2.24%	116.42	116.42	99.20	146.75
IOSI65	65% Fe Fines	100.04	-0.22	-0.22%	130.60	130.60	115.45	171.65

MARKET COMMENTARY

The iron ore market continued its downward trend, with the most-traded contract I2501 closing at 658.5, down 4.5% for the day. Traders showed a low willingness to sell; steel mills had more inquiries, mainly restocking as needed. The market atmosphere was average today. In Shandong, PB fines traded at around 665-670 yuan/mt, down 25-30 yuan/mt WoW from last Friday; in Tangshan, PB fines traded at 685-690 yuan/mt, down 15-30 yuan/mt WoW from last Friday. SMM reported global iron ore total shipments at 35.53 million mt, up 14.4% WoW. Shipments from Australia and Brazil increased, as did those from non-mainstream mines. However, due to low ore prices, the shipment increase from non-mainstream mines was relatively small. The significant supply increase negatively impacted ore prices, combined with inventory at high levels at ports, putting further pressure on prices. Considering pig iron is in a recovery phase and pre-holiday restocking is boosting overall iron ore demand, there is some support for ore prices, and the short-term downside is expected to be limited.

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Sep 20th, 2024		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	820	-20	-2.4%	994	1041	900	1210	103.07	-2.37	-2.25%	126.95	136.27	113.28	153.57

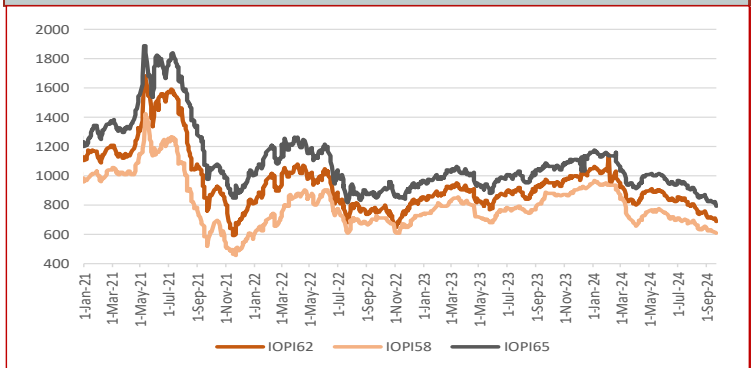
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Sep 20th, 2024				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	870	-0.3%	779	1645	122.83	0.05%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	965	-1.0%	780	1630	136.24	-0.64%	108.71	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	770	-1.3%	620	1310	108.71	-0.89%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	915	-0.4%	800	1752	129.18	-0.04%	0.00	272.32

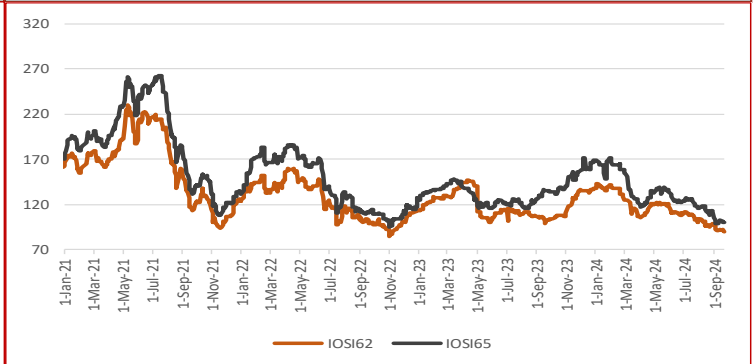
Week Ending Sep 20th, 2024	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	854.57	-0.53%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.7522 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD =6.87528

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Sep 23rd, 2024		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	May	June	July	August	MTD	QTD	YTD	May	June	July	August	MTD	QTD	YTD
IOPI62	62% Fe Fines	894	842	829	765	880	874	895	117.28	110.03	107.99	99.33	115.32	115.83	120.56
IOPI58	58% Fe Fines	762	719	692	656	785	779	781	100.47	94.46	90.48	85.58	103.33	103.96	105.35
IOPI65	65% Fe Fines	1006	954	940	876	996	992	1026	132.42	125.13	123.05	114.39	131.20	131.91	139.19

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Sep 23rd, 2024		CFR Qingdao, USD/dry tonne						
Index	Fe Content	May	June	July	August	MTD	QTD	YTD
IOSI62	62% Fe Fines	120.10	110.42	107.58	99.33	112.14	115.18	116.42
IOSI65	65% Fe Fines	136.06	125.14	123.01	114.06	122.86	129.04	130.60

FREIGHT RATES

Sep 20th, 2024		FREIGHT RATES - DRY BULK US$/wet tonne				
Route	Designation		Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	11.69	-0.20	-1.68%	3.57	16.77
Tubarao - Qingdao	C3	28.04	-0.19	-0.67%	6.70	36.40

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Sep 20th, 2024		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	May	June	July	August	MTD	QTD	YTD	May	June	July	August	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	1024	1011	1001	911	933	989	1017	129.18	127.27	125.75	114.10	119.96	126.47	134.98

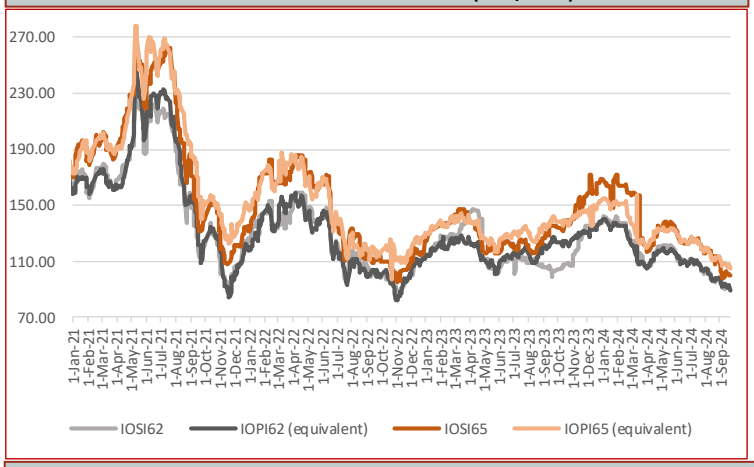
IRON ORE INDEX PREMIUMS/DISCOUNTS

Sep 23rd, 2024		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-73	-10.69%
IOPI65	65% Fe Fines	111	16.25%

Sep 23rd, 2024		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	10.25	11.42%

COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED

IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Sep 23rd, 2024	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	626	-24	-57
SIMEC Fines	551	-25	-132
PB Fines	653	-24	-30
Newman Fines	650	-24	-33
MAC Fines	632	-24	-51
Jimblebar Blended Fines	546	-24	-137
Carajas Fines	863	-24	180
Brazilian SSF	642	-25	-41
Brazilian Blend Fines	666	-24	-17
RTX Fines	565	-24	-118
West Pilbara Fines	595	-24	-88

Sep 23rd, 2024	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	85.30	-2.05	-4.49
SIMEC Fines	81.80	-2.05	-7.99
PB Fines	86.05	-2.05	-3.74
Newman Fines	88.95	-2.02	-0.84
MAC Fines	86.05	-2.05	-3.74
Jimblebar Blended Fines	78.45	-2.02	-11.34
Carajas Fines	119.35	-2.05	29.56
Brazilian SSF	93.55	-2.05	3.76
Brazilian Blend Fines	95.15	-2.07	5.36
RTX Fines	79.65	-2.07	-10.14
West Pilbara Fines	84.05	-2.05	-5.74

Sep 23rd, 2024	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	610	-7	0
FMG Blended Fines	618	-7	8
Robe River	618	-7	8
Western Fines	621	-7	11
Atlas Fines	616	-7	6
Yandi	604	-7	-6

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	5.00	0.00
	High Grade Fe 63 - 64%	5.00	0.00
	High Grade Fe 64 - 65%	5.00	0.00
	High Grade Fe 65 - 65.5%	5.00	0.00
	Low Grade Fe	26.00	0.00
1% Alumina	High Fe Grade Al <2.25%	60.00	0.00
	High Fe Grade Al 2.25-4%	5.00	0.00
	Low Fe Grade Al <2.25%	40.00	0.00
	Low Fe Grade Al 2.25-4%	25.00	-4.00
1% Silica	High Fe Grade Si <4%	37.00	1.00
	High Fe Grade Si 4-6.5%	9.00	0.00
	Low Fe Grade	37.00	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	8.00	1.00
	High Fe Grade 0.115%<P<0.15%	8.00	1.00
	Low Fe Grade 0.09<P<0.1%	5.00	0.00

Seaborne Index Product Differentials (USD/dry tonne)

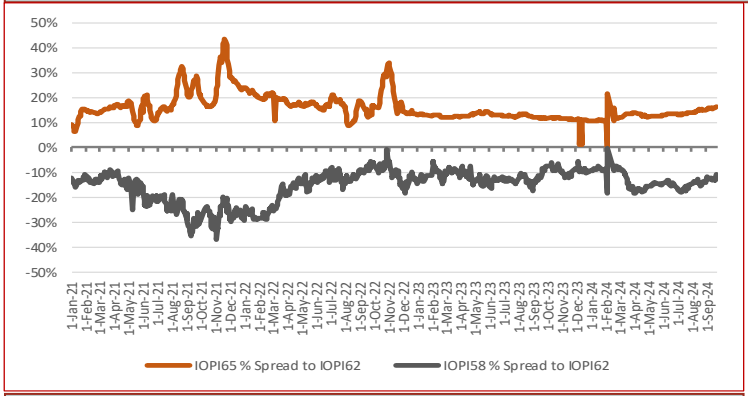
	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	3.50	0.00
	High Grade Fe 63 - 64%	0.25	-0.75
	High Grade Fe 64 - 65%	0.25	-0.75
	High Grade Fe 65 - 65.5%	0.25	-0.75
1% Alumina	High Fe Grade Al <2.25%	0.25	0.00
	High Fe Grade Al 2.25-4%	1.75	0.75
1% Silica	High Fe Grade Si <4%	0.50	0.25
	High Fe Grade Si 4 - 6.5%	0.50	0.25
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	2.00	0.25
	High Fe Grade 0.115%<P<0.15%	2.75	0.25

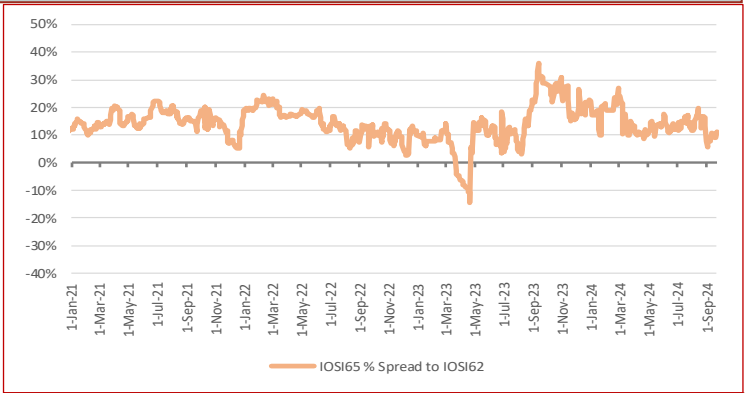
Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-50.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-30.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	-15.00	0.00	Jingtang	-15.00	0.00	Majishan	0.00	0.00	Taicang	-20.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	-10.00	0.00

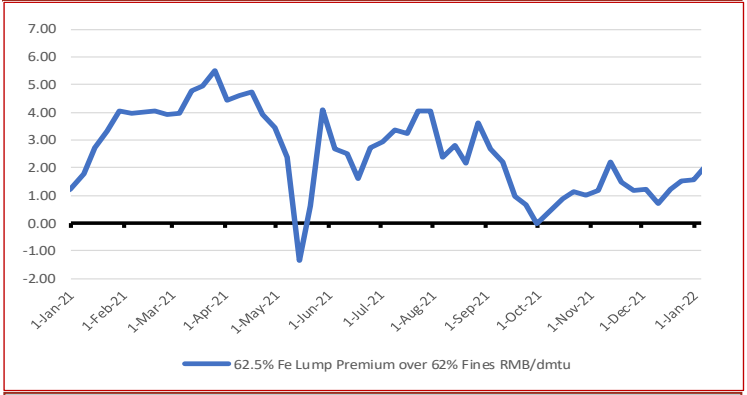
COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED

IRON ORE INDEX PREMIUMS/DISCOUNTS

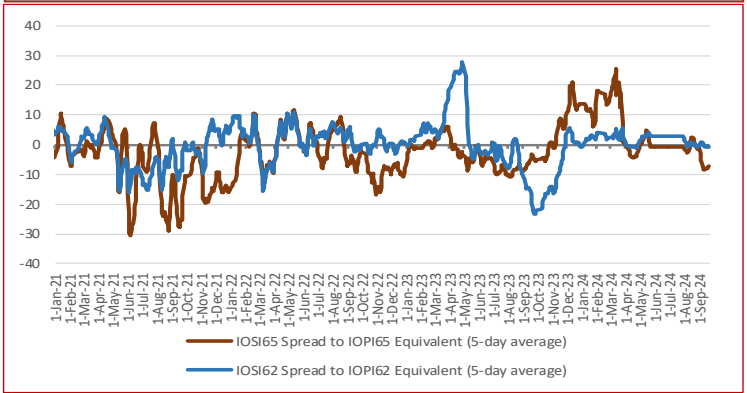




WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)

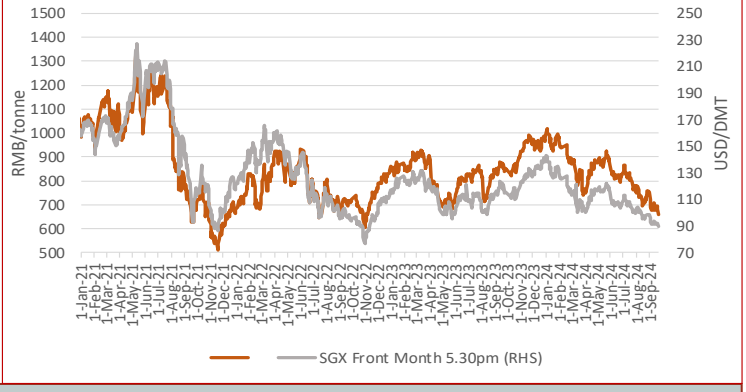


IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)

FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

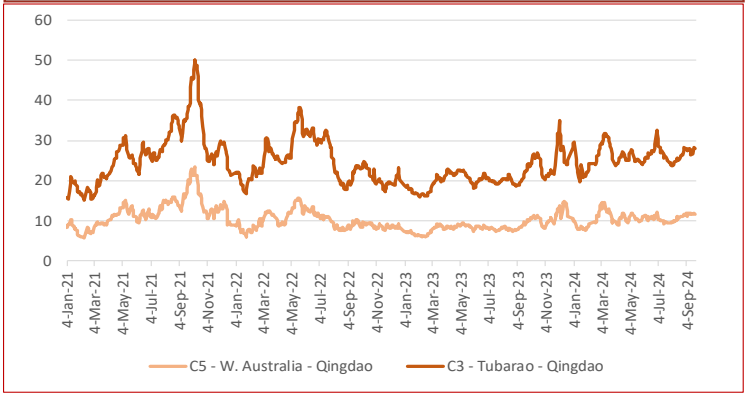
Week Ending Sep 20th, 2024

Province	This week	Change %	Low [2]	High [2]
Jingtang	14.27	-2.19%	8.41	19.20
Qingdao	24.75	-1.59%	9.41	26.24
Caofeidian	17.74	-2.58%	7.56	18.47
Tianjin	11.96	-0.83%	6.49	12.97
Rizhao	20.99	1.30%	9.44	20.99
Total (35 Ports)	150.00	-0.48%	98.80	155.39

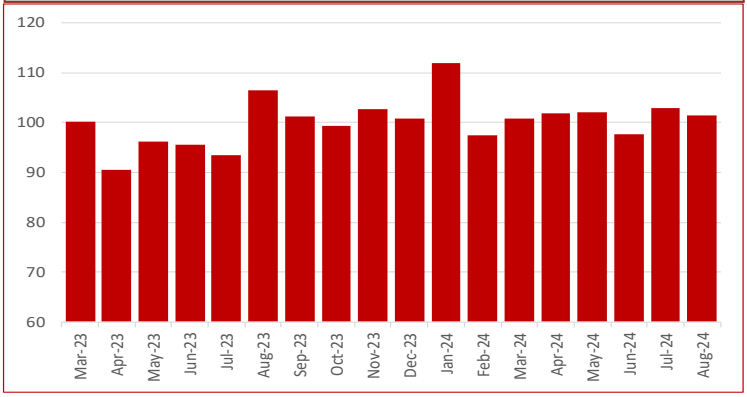
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Sep 23rd, 3pm close			Sep 23rd, 5:30pm		
Contract	I2501	Change	Change %	October. 24	Change	Change %
Closing Price	658.50	-21.50	-3.16%	89.45	-2.70	-2.93%
Vol traded ('000 lots)	60.20	-2.98	-4.71%	12.03	2.43	25.34%
Open positions ('000 lots)	62.50	4.95	8.60%	38.67	0.51	1.33%
Day Low	658.0	-18.50	-2.73%	89.20	-2.05	-2.25%
Day High	681.5	-21.50	-3.06%	92.10	-2.15	-2.28%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED

STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2024/9/20	Change	Change %
ReBar HRB400 ϕ18mm	3,310	0	0.00%
Wirerod Q300 ϕ6.5mm	3,550	-10	-0.28%
HRC Q235/SS400 5.5mm*1500*C	3,140	10	0.32%
CRC SPCC/ST12 1.0mm*1250*2500	3,560	10	0.28%
Medium & Heavy Plate Q235B 20mm	3,270	30	0.93%
GI ST02Z 1.0mm*1000*C	3,720	10	0.27%
Colour Coated Plate	6,050	0	0.00%

CHINESE STEEL MILL PROFITABILITY

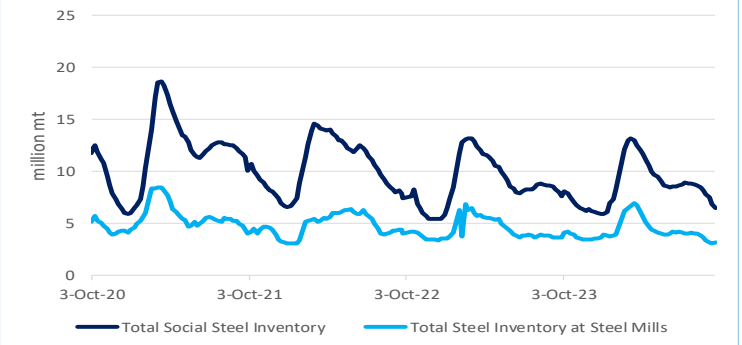
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	91.85	-0	Mmi CFR Equivalent index for 1st Feb
Coke	1,845	55	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	2,360	0	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	2,513	10	Q234, incl. tax
Rebar cost - Blast furnace	3,065	11	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	165	-11	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	3,145	10	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	-15	-10	based on Shanghai prices, incl. tax

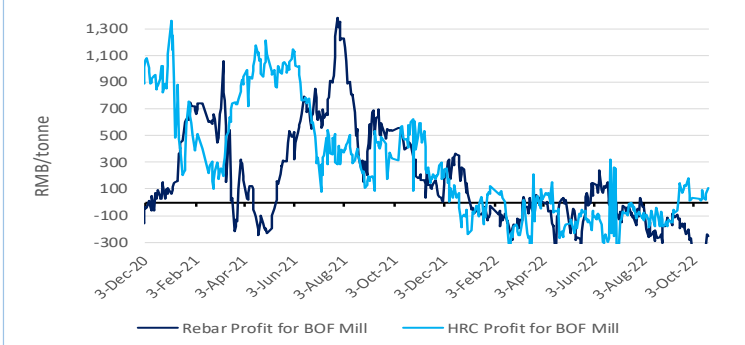
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

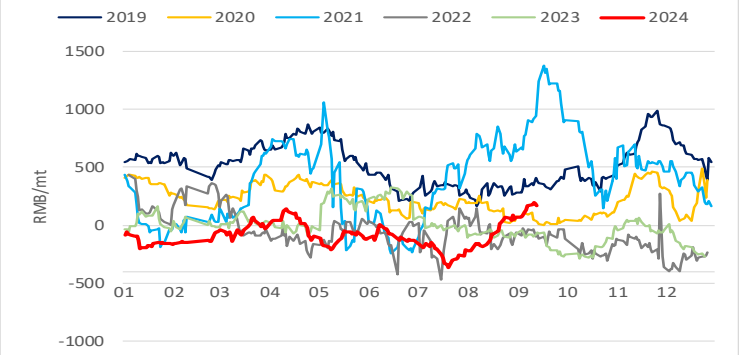
CHINESE STEEL INVENTORIES



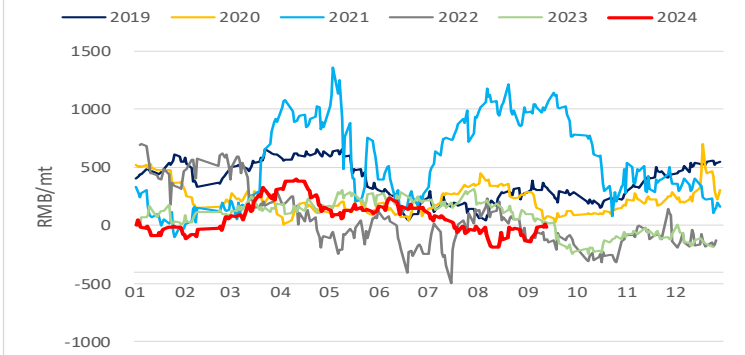
CHINESE STEEL MILL PROFITABILITY



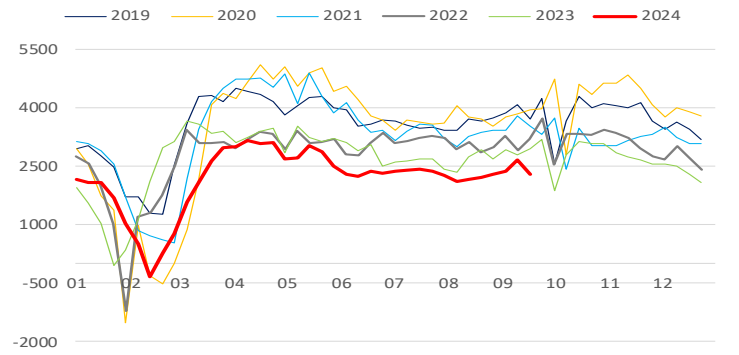
CHINESE STEEL MILL PROFITABILITY—Rebar



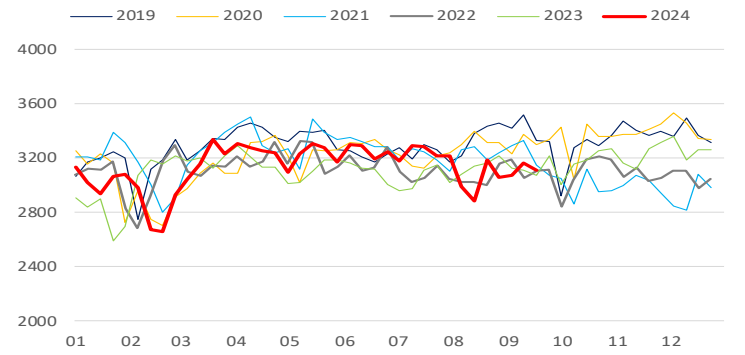
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



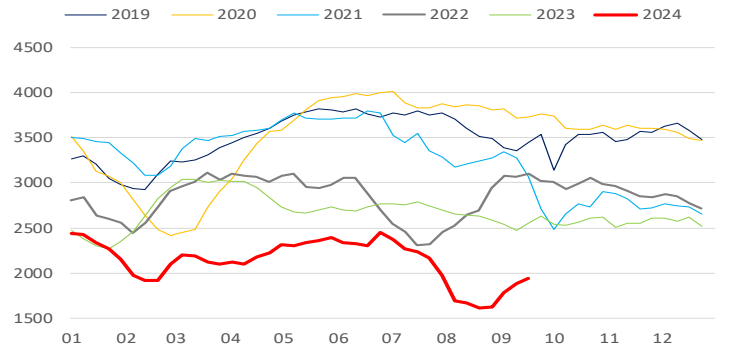
CHINESE STEEL CONSUMPTION—Rebar



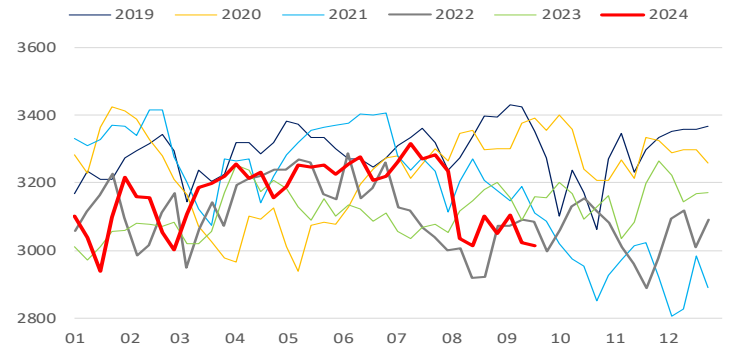
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED

IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Sep 23rd, 2024	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Sep 23rd, 2024	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.30%	0.095%	8.00%
MAC Fines	60.80%	2.35%	4.40%	0.075%	7.80%
Jimblebar Blended Fines	60.50%	3.10%	4.80%	0.120%	7.80%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

CONTACT US

MMI Singapore Office Details: Level 28, Manulife Tower, 8 Cross Street Singapore. Tel: + 65 6850 7629 E: prices@mmiprices.com
SMM Singapore Office Details: Level 28, Manulife Tower, 8 Cross Street Singapore. Tel: + 65 6850 7630 E: service.en@smm.cn
SMM Shanghai Office Details: 9th FL, Building 9, Lujiazui Software Park, No.20, Lane 91, Pudong, Shanghai Tel: +86 021 5155 0306 E: service.en@smm.cn

Visit www.mmiprices.com for full index price histories and archive of daily reports

This information has been prepared by Metals Market Index ("MMi"). Use of the information presented here is at your sole risk, and any content, material and/or data presented or otherwise obtained through your use of the information in this document is at your own discretion and risk and you will be solely responsible for any damage to you personally or your company or organisation or business associates whatsoever which in anyway results from the use, reliance or application of such content material and/or information. Certain data has been obtained from various sources and any copyright existing in such data shall remain the property of the source. Except for the foregoing, MMi retains all copyright within this document. The copying or redistribution of any part of this document without the express written authority of MMi is forbidden.

COPYRIGHT METALS MARKET INDEX, ALL RIGHTS RESERVED